Aditya Naidu

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Education & Certifications

Acropolis Institute of Technology and Research (AITR) Indore, MP
B.Tech in Computer Science & Engineering (AI/ML) *Expected Sep 2027*
Certifications: Quantization Fundamentals (Hugging Face & DeepLearning.AI), AI Agents in LangGraph (LangChain)

Technical Skills

Languages: Python, C++, SQL, HTML, CSS
Frameworks & Tools: ROS2 (Humble), PyTorch, ONNX Runtime (GPU), OpenCV, LangGraph, AutoGen, FastAPI, Git, colcon, WSL2/Ubuntu 22.04, IPG CarMaker
Domains: Machine Learning, Multi-Agent Systems, Computer Vision, Autonomous Vehicles/ADAS, Sensor Fusion, Data Science

Experience

Team Acroracerz (aBAJA SAE India, Vehicle #264001) Indore, MP
Captain & AI/ADAS Lead Developer *Jan 2026 – Aug 2026*
- Led an interdisciplinary engineering team migrating autonomous vehicle software to a distributed ROS2 (Humble) architecture, serving as lead representative for national technical design and safety reviews.
- Engineered custom ROS2 hardware drivers for IMU, GNSS, Hall-effect wheel-speed sensors, and cameras to establish a deterministic, low-latency multi-modal sensor fusion pipeline.
- Deployed Lane Keep Assist (LKA), Autonomous Emergency Braking (AEB), and an Endurance Arbitrator on an NVIDIA Jetson Orin NX, achieving deterministic 50 Hz closed-loop drive-by-wire control.
- Validated perception and control stacks in IPG CarMaker simulation and physical track testing, driving the vehicle to All India Rank 1 in AEB and the Drive-by-Wire Award among 60+ collegiate teams.

Projects

Multi-Head Agent | *Python, LoRA, LLMs* | [GitHub] *2026*
- Architected and implemented a multi-agent system using a frozen shared backbone LLM equipped with role-specific LoRA adapter heads for Planner, Critic, and Solver roles.
- Integrated a shared KV-cache and latent communication hand-offs between adapter heads to eliminate redundant token generation and replace slow text-based inter-agent hand-offs.

Palette: Multi-Agent Orchestration Framework | *Python, LangGraph* | [GitHub] *2025*
- Built a modular Python framework on LangGraph that abstracts multi-agent workflow boilerplate, cutting pipeline setup time by 70–80%.
- Implemented a round-robin StateGraph with Primary and Critic validation agents, conditional-edge termination, and per-agent switchable LLM providers.

Edge Vision Super-Resolution Engine | *PyTorch, ONNX Runtime, OpenCV, C++* | [GitHub] *2026*
- Implemented an efficient sub-pixel CNN (ESPCN) in PyTorch to upscale 360p video streams to 1080p in real time, computing feature maps in low-resolution space to eliminate throughput bottlenecks.
- Applied post-training INT8 quantization and operator fusion via ONNX Runtime, reducing model size by over 70% while maintaining real-time frame rates on low-power edge and mobile hardware.

Honors & Awards

aBAJA SAE India 2026 (Phase 3, GARC Chennai): All India Rank 1 (AEB Event), Drive-by-Wire Event Winner, and Overall 2nd Runner-Up out of 60+ national collegiate teams.

Smart India Hackathon (Nov 2025): College Round Winner – Engineered a Raspberry Pi 5 edge device running a dual-engine OCR pipeline (EasyOCR/Tesseract) and FastAPI backend for automated packaging compliance.

HackIndia (May 2025): National Finalist – Built an automated meeting summarization pipeline using multi-agent orchestration in Microsoft AutoGen.

Kryeta Hackathon (Apr 2025): Finalist – Prototyped a multi-agent orchestration architecture using Microsoft AutoGen with iterative critic validation loops.